

A-Team Tutorials

Level 3

Targets and Priorities

Asteroids

NEOFixer is the main source of NEO observing priorities. We should also consider unconfirmed objects and various other targets.

Near Earth Object Confirmation Page (NEOCP)

The [Near Earth Objects Confirmation Page](#) contains objects that have been recently discovered and require confirmation. Its focus is on objects that could be NEOs. Each object on this page has a “score” indicating how likely it is to be a NEO based on the observations so far.

The NEO Confirmation Page

Please ensure you are familiar with the [notes at the bottom of this page](#).

Page last updated on Sept. 4.582 UTC

Problems? Comments?

Get ephemerides Reset form

Select object(s) from the current list of objects needing confirmation (NEO desirability score, discovery date, rough current position and magnitude given, as well as number of observations, arc, nominal H and number of days since it was last observed):

☐ All objects with $V =$ to , with Decl. between -90° and $+90^\circ$, with an NEO desirability score of % to %

☒ or just the objects selected below:

Temp Desig	Score	Discovery	R.A.	Decl.	V	Updated	Note	NObs	Arc	H	Not Seen/dy
☐ COWDX22	100	2019 09 04.5	04 57.2	+01 01	21.4	Added Sept. 4.52 UT		4	0.02	27.7	0.079
☐ COWDWF2	76	2019 09 04.5	04 53.4	+04 20	20.8	Added Sept. 4.51 UT		4	0.02	16.7	0.081
☐ COWDTN2	100	2019 09 04.5	04 57.7	+05 41	20.4	Added Sept. 4.51 UT		4	0.02	23.1	0.081
☐ A10FWHj	100	2019 09 04.4	22 26.6	-14 47	19.2	Added Sept. 4.51 UT		4	0.02	24.1	0.147

NEOCP webpage

Objects on the list have Temporary Designations assigned by the discoverer. When sufficient observations have been acquired to confirm the existence of an object and generate an initial orbit, the MPC assigns a Provisional Designation (which it retains until it becomes a Numbered Object) and publishes a Minor Planet Electronic Circular (MPEC) announcing it.

NEOCPs are usually desirable targets but many are too faint, have only a handful of observations and uncertain orbits. Bill Gray (developer of Find Orb) provides a [summary of NEOCP objects](#) that gives additional information derived from Find Orb automated processing. This can help identify viable targets.

When an object disappears from the NEOCP it may be found, with its new Provisional Designation, at:

[Previous NEOCP Objects](#)

This list also carries comments like “was not confirmed” or “was not a minor planet” for objects that were never confirmed, were lost, or turned out to be a satellite.

The ESA provides similar data in its [NEOSCAN](#) page.

Potential Mission Destinations

A selection of objects has been designated potential destinations in the Near-Earth Object Human Space Flight Accessible Targets Study (NHATS). There is a customisable list of these objects at:

[NHATS](#).

Lowell Observatory

Various lists and query functions are provided at [Asteroid Observing Services](#).

Sentry Risk Table

The NASA [Sentry system](#) monitors the probability of Earth impact of NEOs. All risk-related objects are listed but there are no facilities to select observable ones.

The [NEODyS website](#) lists the currently observable risk-related objects.

If a new object is discovered and is considered to have an impact risk, it is placed on the Sentry list, and the status of “Virtual Impactor (VI)” will appear against it in the MPC Ephemeris pages. It will also appear as urgent in priority lists.

Approachers

Regardless of urgency, objects making a close approach are often interesting and in the news. ESA publishes lists of “[forthcoming and recent](#)” objects approaching the region inside 0.05 AU of Earth.

Comets

At Level 3 we include comets in the candidates. Although the MPC maintains a database of comet observations there is much less emphasis on providing observation priority or guidance. If a comet is considered a Near Earth Object, it will appear in NEOFixer listings the same as an asteroid.

Observable comets

The MPC Database includes observations and orbits of comets, and the Ephemeris page will provide coordinates for comets. Observation planning aids are limited, but there is a [list of observable comets](#) and a list of [dates-of-last-observations](#).

A convenient way to look for comet candidates is to use your planetarium software. Orbital Elements for currently observable comets, suitable for loading into planetarium programs, can be found [here](#).

Seiichi Yoshida's Comet Lists

Seiichi Yoshida maintains two useful lists that often contain better magnitude estimates and are more “user friendly” than the MPC comet list:

[Weekly Information about Bright Comets](#) - visible comets.

It is useful to observe these comets from time to time to check their position but more importantly to check their magnitude and general development.

[Comets Waiting for First Observation](#) – Recovery opportunities.

These are periodic comets that should be visible again soon. There is kudos to be gained by being the first to report a sighting. Because of the effects of outgassing, the positions of these objects may be uncertain, and it is unpredictable when they will begin to develop sufficient coma to become visible. Recovery may be challenging.

Possible Comet Confirmation Page (PCCP)

If a newly discovered object on the NEOCP appears “fuzzy”, the discoverer/observer should indicate this fact using an [online report form](#).

Initially, the MPC may not publish this comment in order not to influence other observers. If further observers also comment that it looks like a comet then it will move to the [Possible Comet Confirmation Page](#) (PCCP).

Here it will remain until fully confirmed and an initial orbit established. It will then receive a comet designation and an MPEC will be published to announce it. Follow-up observations of these comets are always worthwhile. Some objects flip between the PCCP and NEOCP as observers are unsure if they can see cometary activity (coma, tails). If you report cometary activity, your report will be included in the MPEC, but do not report a coma or tail unless you are sure. People who make false reports go on a blacklist!

Harvard ICQ/EPS

The Harvard International Comet Quarterly/Earth and Planetary Sciences website has various links and lists that may be helpful if you are looking for information about a particular comet.

[International Comet Quarterly](#)

Discoveries

There is, of course, the alternative of trying to discover new objects. This is a highly competitive sport, especially with several well-equipped professional survey teams continually scanning the sky. There is one source of assistance at the [Sky Coverage Plot](#). This MPC tool generates a map of where the major surveys have been imaging recently. It can be a starting place to find where not to look!

Both Astrometrica and Tycho Tracker have functions to search for moving objects in a set of images.

Visibility and Missions

Uncertainty

You are already experienced in finding coordinates and setting up missions for most objects, but we need to consider the situation when the coordinates are uncertain.

NEOCs and PCCPs may have very few observations, and their ephemeris may have an uncertainty that renders them difficult to observe. Some NEOs have few observations from years ago. You need to obtain the best possible coordinates for your missions and may need to plan several missions to search for an object.

The MPC Ephemeris page may include specific uncertainty information and an “uncertainty map” but this can be unreliable. Be particularly wary if you see a “One Opposition” object with no uncertainty map. It is probably “lost”.

Find_Orb Ephemeris

A good way to get ephemeris and uncertainty information is to use the ephemeris-generating function of Find_Orb. Details of how to do this are described in the Find_Orb tutorial. Unfortunately Find_Orb is a Windows application but MAC-users with Tycho Tracker have a limited-function version attached to the Session Planning modules that “may” produce useful ephemeris.

Fast movers

The 20-second Multi-luminance recipe can be used to observe faster moving objects provided they are bright enough to provide sufficient SNR with the short exposures. The timestamps on images from Slooh telescopes are generally accurate to ± 0.1 second. This translates to an error of $\pm 0.25''$ for an object moving at $150''/m$. So, in theory, a reasonable measurement could be made of an object moving at that speed.

However, such an object would cover a distance of $50''$ during a 20s exposure. So, the challenge is more related to accuracy of measuring a long streak rather than timing precision.

For extremely bright objects, you can try the photometric 10s or 4s recipes.

Identify and Measure

The basics of data reduction and measurement have already been covered in previous levels but here we need to look at some more difficult situations.

Close to Star

If your target is close to a bright star, the centroid may be dragged towards that star and give wrong position and magnitude.

- With Astrometrica you can use the Centroid button to examine the situation and find an aperture that will fit the target and avoid most of the star. Go into settings and change the aperture to that size (do not save!). Then try to obtain a measurement using the small aperture. Set the “close to star” note for this observation and make sure to check residuals.
- With Tycho Tracker you can use the zoom, contrast and intensity sliders of the Image Viewer to try and isolate the target from the bright star. Double-click on the target to see if you can get it into the centre of the cross-hairs then Create/Observation.

Crowded field

If your target has a lot of background stars it may be difficult to isolate it against all the star trails. Visibility can sometimes be improved by using a Median stack rather than an average. Some of the trails will be removed or thinned out.

- In Astrometrica, select the median stack method when setting up Track & Stack.
- In Tycho Tracker, the stacking method can be changed in the Image Viewer.

You can also examine your individual images and select just those where your target is clear of stars.

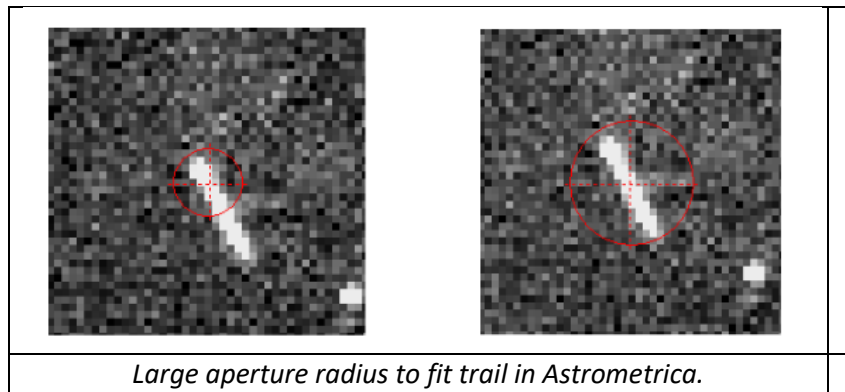
Fast movers

If an object is moving fast, it may produce a streak or trail in the image. Short streaks of up to 4 or 5 pixels may not cause a problem but with longer tracks it may be difficult to create an observation accurately based on the centre of the track.

If in doubt, try your normal way of generating observations. See if it “looks like” the observation is in the centre of the trail

Check the reported PSF curve (Object Verification dialog in Astrometrica, View PSF in Tycho Tracker) and examine the way in which the centroid has positioned itself. Has it clearly looked for the centre of the streak in terms both of width and length?

- Using Astrometrica, you can try to enlarge the Aperture Radius until the trail just fits inside, and the measurement is in the centre of the streak.
- Using Tycho Tracker you can create Markers that just fit over the ends of the trail, then use Create Observations for Streaked Object (in the Markers dialog). This produces 3 observations. One for each end of the streak and one for the middle. Use the middle one.



Reference stars problems

If your coordinates are in some regions of the sky, there may be too few stars for the software to use as references. In other areas there may be so many stars that the software gets confused and data reduction/plate solving fails.

To handle these situations be aware of the various catalogs that can be used and the Settings of Upper and Lower magnitude limit that can be applied. The minimum and maximum magnitude limits in catalog settings (Astrometrica or Tycho Tracker) can be used to increase/decrease the number of stars being used.

However, make sure the lower magnitude setting does not let saturated stars be used for reference.

Focal length and rotation

Astrometrica relies on the Settings having correct values for focal length and field rotation. After a servicing visit (or extreme temperatures) these figures can change or drift.

Changes to focal length or rotation are periodically mentioned in the A-Team Club but if you are having trouble getting your images to register during data reduction, look in the log file of Astrometrica (File/View Log File.). Find where it states the focal length and field

rotation for a successfully registered image and use those values in the settings.

Tycho tracker does not normally suffer from this problem as it uses a more flexible plate solver.

Stacking Uncertain Objects

Normally, when doing Stack and Track you get the Motion and PA from the Ephemeris or it is derived from the object's orbit in MPCORB. However, NEOCPs and new discoveries are not in MPCORB and the motion/PA data for objects with a high positional uncertainty may be quite wrong. In these cases:

- Use Find Orb with whatever observations are available to generate an ephemeris with motion and PA.
- Run Stack & Track with this motion and PA.
- Make a set of measurements, add these to the old observations, re-run Find Orb and make new ephemeris.
- If there is a big difference between your original motion and the that shown in the new ephemeris, repeat Stack & Track using the new figures and remeasure the object.

'Known Objects Box' problems

There are several reasons why an object may not appear in its known-objects box or centred in the cross hairs.

- An object's orbital elements may have uncertainty. If the uncertainty is moderate the object will be seen (when blinking) moving close to its indicated position. This is perfectly normal.
 - Astrometrica: Measure and report its position as usual.
 - Tycho Tracker: Double-click on it to put it in the centre.
- Sometimes you will see indicators for objects that should be bright enough to be visible – but no object in the image. Check its uncertainty in the MPC Ephemeris and Observation DB. If it had only a handful of observations a few years ago, it is probably lost and could be anywhere. The position derived from MPCORB is wrong.
- If your copy of MPCORB is old, it will not include recently discovered objects. You will need to update MPCORB. In Tycho Tracker you can set this to happen regularly but in Astrometrica you have to do it manually.
- If you are processing an old image, then the epoch of a recent MPCORB may be a long way from the date of the image and positions derived from it may be way out.

In the case of a recently discovered NEO undergoing a close approach, downloading a new copy of MPCORB may not provide an accurately positioned box due to the way in which the MPC only recalculates orbits periodically. You can fix this with Internet/Update MPCORB/Near Earth Objects in Astrometrica (but this is quite slow). Using Tycho you should get the orbit from JPL.

NEOCs

For NEOCP objects proceed as usual and request Known Objects. This is to identify the known objects in the image and not mistake one of them for the NEOCP.

The NEOCP itself will not have a Known Objects indicator (because it is not a known object!). You will need to be exceptionally careful to check you have the right object and that it is not a false alarm (see below). The target may be some distance from the coordinates originally computed especially if there are not many prior observations.

- Astrometrica: In the Object Verification dialog enter the NEOCP designation in the Object Designation box. Be sure to enter it exactly as shown on the NEOCP page.
- Tycho Tracker: Enter the NEOCP designation in the Provisional Id of the Object Designation dialog.

Check your residuals in the normal way to ensure your observations fit properly with the existing ones.

It is as well to check the NEOCP page at the MPC to see that the object is still there. If it has been promoted to a designated object you will need enter its new Provisional Designation instead.

You can find a list of newly assigned designations [here](#).

Comets

When the target is a comet with no visible coma or tail then it may look just like an asteroid in which case you can process it in exactly the same way. If it has a tail and/or coma then some new techniques are needed.

What's the problem with comets?

Comet images may be fuzzy and not symmetrical about their true nucleus. Measurement software has problems fitting a PSF curve to them.

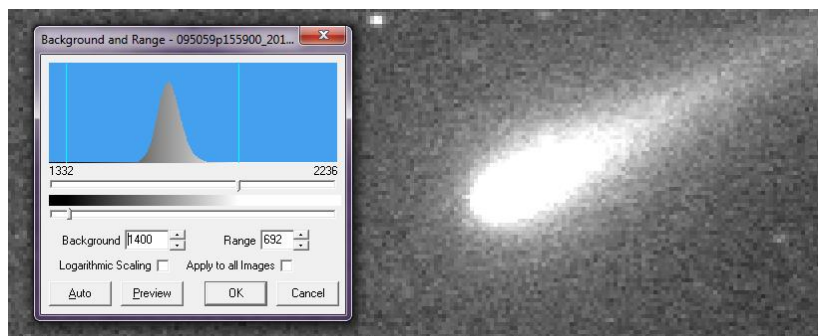
Comets may be influenced by non-gravitational forces caused by outgassing. Their orbits may not be exactly as expected.

Finding the nucleus

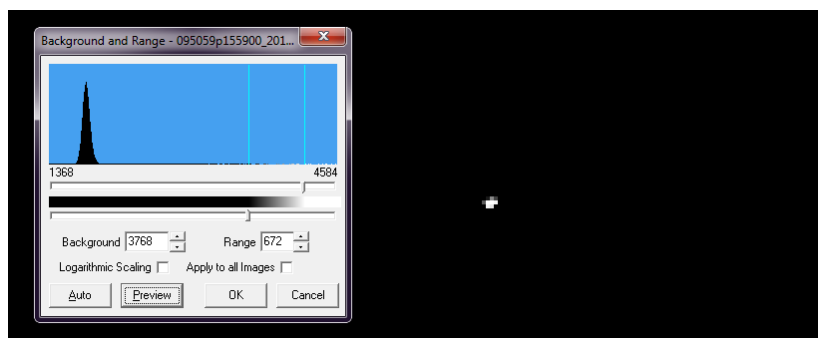
The position of a comet is the position of its nucleus. The nucleus has the mass and determines the orbit. To judge the position of the nucleus, look for a small cluster of the brightest pixels, and measure its position.

Astrometrica:

After data reduction (and track & stack if needed) select one of the images then “Images/Background and Range”. Use the counter to increase the Background value, periodically pressing “Preview” until the brightest point in the coma shows up as a small cluster of pixels. Do not press “OK” during this process as that will reset “Range” which we do not want.



Comet as initially shown



Background adjusted to highlight the nucleus

Select “Apply to all Images” and OK when you have the desired result.

In Settings/Program set Aperture radius to a small number. Typically, 2 or 3 will be appropriate depending on the telescope system used. Sometimes the comet has a very diffuse centre, and slightly bigger aperture may be needed. In any case, the centroid should fit snugly round the very brightest pixel or small cluster of pixels in the head. Do not save this change to the configuration file!

Click carefully on the centre of the bright cluster of pixels, (use the magnifier and remember you can use the arrow keys and Enter key to move and set the position). The PSF curve will not fit the distribution of pixel values as neatly as it usually does for asteroids, but it should at least find a “hump”. If there is no hump, then the

PSF method will not work. In that case press the Centroid button and adjust the position for maximum flux. Set object identification and Accept in the normal way.

Tycho Tracker:

The same effect can be obtained in Tycho Tracker. Process the images as normal. When you have the comet displayed in the Image Viewer, increase the contrast and reduce intensity sliders so the only a small cluster of pixels remain visible. Double click the cluster to place it in the centre of the cross hairs then Create/Observation.

Discoveries

Moving Object Detection

If your objective is to discover new objects, then proceed as normal with data reduction and display of known objects.

The simple approach is to blink the images and study them carefully. You will immediately “see” lots of things apparently moving about.

You can try using Astrometrica’s Moving Object Detection tool – it is not very sensitive but may come up with something. The tool works with a minimum of 3 and a maximum of 6 images.

Tycho Tracker has a much more powerful tool called Synthetic Tracker. The images are repeatedly stacked and tracked with different motion a position angle then searched for objects. Quite faint objects can be detected. The downsides are that you need a good GPU to make it work and many false detections are made.

False alarms

Most, (usually all) of the apparently moving objects in the image will be false alarms.

- Hot pixels are pixels that have “stuck” with the maximum value and appear as single pixel-sized points of brightness. They may appear to move from image to image because the telescope pointing has moved. Under magnification the brightness has a sharp edge, not like a real object that is smoothly spread by the PSF. However, there is a danger that hot pixels get to look like real objects after tracking and stacking multiple images.
- Cosmic Ray (CR) hits are individual or small groups of pixels that have been brightened by the impact of a cosmic ray particle. The extent of the brightness varies according to the particle energy and its direction of impact. The bright region usually has sharp edges that do not look like a real object spread by the PSF. An individual CR hit will not “move” but it is easy for your eyes to deceive you into believing that different CR hits on

consecutive images come from a moving object. There are lots of CR hits in most images.

- Satellites can sometimes appear as moving objects. Most move quickly and will be obvious. There are some satellites in higher orbits that regularly get “rediscovered” and appear for a while in the NEOCP. Subscribe to the [groups.io MPML](https://groups.io/g/MPML) where these objects are often discussed.
- Beware also of internal reflections (e.g., from Moon light), dust motes (donuts), spots of condensation (usually the domes are closed before that can happen) and ghost images (typically Jupiter from the previous mission).

Temporary Designations

If you still think you have found something you will need to invent a temporary designation for it. Typically use your initials followed by a sequence number. (The professional surveys use complicated structures as seen in the NEOCP designations). The main rule for temporary designations is that they must not look like the name, number, or designation (full or packed) of a known object.

Measure the object in the normal way putting your temporary designation in the Object Designation field.

Use the [MPCChecker](#) to make sure it is not already known. It may be worth checking the NEOCP list to see if someone else has reported it. You may also want to use the [NEOChecker](#) to get a score for it and decide if you want to submit it as a new NEO.

A minimum of 3 (preferably 4) images, showing the object moving with consistent speed and direction, should be obtained before believing you have found something.

Check the residuals to ensure the discovery is moving in a straight line with consistent motion.

Reports and Publications

At Level 3 there is no separate tutorial on Residuals and Quality. You have Find_Orb and it is up to you to decide whether you think your observations are good enough to submit. This applies to asteroids, comets, close approaches, NEOCPs etc.

However, be aware that comets sometimes suffer from non-gravitational forces (from jets of gas issuing from their surface). The Find_orb application can take these forces into account, but the on-line and basic Tycho Tracker version cannot. This can lead to high residuals even for good observations.

A-Team Report

As in the previous levels, please post a report to the tutor or A-Team Club containing:

- A description of the target and why it is important.
- Comments about any problems with imaging or measuring.
- The .psv file you propose to submit.
- Find Orb residuals of the observations you propose to submit.

You may like to post an animated gif of the observations in the A-Team Club as do several other A-Team members.

MPC Report

The basic structure of the MPC report has been discussed before. The MPC Submissions page has some additional indicators. Do not mix up different kinds of objects in the same submission.

- NEOCP – Observations of an object on the NEOCP/PCCP.
- New NEO Candidate – Check this if you think your observations are the discovery of a new NEO.
- NEO – Observations of a known, designated, NEO.
- New Comet – Check this if you have discovered a new comet.
- Comet – Observations of an existing comet.
- TNO – Observations of a distant object.
- ARTSAT – Observations of a known Artificial Satellite.

The MPC will respond to a NEOCP report, usually within a minute or two, confirming the identity of the object. Your observations will appear in the observations list also within a few minutes. Your observation and name will appear in the announcement MPEC if the object is confirmed as a NEO.

If your target is a PCCP or you believe you can see cometary features, then you should use the MPCs special report page in addition to the normal ADES submission.

<https://minorplanetcenter.net/mpcops/submissions/cometary/>

The form is self-explanatory, and you must use it if you want your comments to appear in the MPEC. You must not use it for any known (designated) comet.

Discoveries

A potential discovery raises the question as to whether to report it immediately (and you may be wrong) or wait for a confirmation (and someone else may report first). There are some considerations:

- Are you sure it is a moving object and not an image artefact?
- Has a survey recently checked this region of sky? If so, is it reasonable that you have spotted something they missed?
- Have you used the [MPC Checker](#), [Comet Checker](#) and/or [Distant Satellite Checker](#) to check for a known object?

Use the [NEO Rating tool](#) at the MPC to obtain a “score” showing how likely it is that your object is a NEO. If it has a score of more than 50 then mark it as “New NEO Candidate” in the submission form.

At the end of the day, if you are feeling reasonably confident, send the report and see what happens.

Try to follow up as quickly as possible. Use Find_Orb to predict where it will be tomorrow night or later today at another Slooh telescope.

Note you will not be officially credited with being the “discoverer” until the object becomes numbered, and it is confirmed that you provided the earliest observations as defined in [MPEC 2010-U20](#). Sadly, it is very unlikely these days that you will get credit because the big surveys, with years of images, can probably show they saw it first. There is more information [here](#).

Submissions

If approved by your tutor, send the report to the MPC.

Once you have demonstrated the ability to execute the complete workflow reliably a few times, with a range of object types and good judgement regarding your residuals, you will “graduate” and be authorised to send reports without needing to check with anyone. However, I hope you will continue as an active member of the A-Team:

- Post information in advance about your targets so we don’t duplicate our efforts.
- Post results and experiences because we are all interested in seeing who is observing what.
- Occasionally post a gif and report as an “Observation” so that all members can see what we are up to.

Publications

[MPC Publications](#) remains the best places to look for your observations. You can also find them in the [NEODyS](#) or [AstDyS](#) systems with their residuals and indicators as to whether they were used as part of the orbital solution.

ADS (SAO/NASA Astrophysics Data System)

ADS is a formal bibliographic reference and abstract database for matters relating to astronomy and physics. The checks made by the MPC constitute “peer review” of your observations and after publication they are eligible for inclusion in the ADS service. Entry to the service is at:

[ADS Query Form](#)

You can search using your name (last name, first initial) and in the title field put “Observations and orbits of comets”.

Congratulations! You are now a named (co)author of a formal scientific document! Note, however that it may take some time before your name (as co-author of an MPEC) begins to appear in the ADS listings.

Conclusion

This is the end of the A-Team Tutorials. I hope it was challenging but enjoyable.

Congratulations on getting this far!

But now – we need your help to improve these tutorials for the next generation of students. Please provide comments and suggestions and any additional notes about special techniques you know.

Now you are a fully-fledged A-Team member, and you can research topics of your own. Have fun!

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Tony Evans